

Faculty of Computing

Assignment – Mini Project

IT1140 – Fundamental of Computing

Year 1, Semester 1

Assignment released	: 1 st Week
Due date	: 12 th Week
Total marks	: 100
Deliverables	: Proposal document (10 marks) Proposal presentation (10 marks) Progress presentation (20 marks) Final Assessment/ Demonstration (60 marks)

This is a group assignment with six members in each. Imagine that your team has started a startup company that is expecting to develop an embedded (combining computer hardware and software) product for the consumer market. With a brief technical study, you must select a device that will fulfill the following functions.

- One or more sensors as input.
- An actuator or an indicator as an output.
- Use of memory to save relevant data.

The market is always expecting something amazing. Hence, if you are capable, you can add the following to your product.

- A web interfaces.
- A mobile app.

Once the prototype is completed, select a problem from the following.

- A water tank monitoring system
- A home automation system
- Automated health care monitoring system
- A learning tool for children
- A greenhouse automated system

Then, you must identify the scope of the problem that you are going to address and sort out the functions.

Next, you are supposed to design the solution (a prototype as a tangible product) for the problem. As the project is an embedded system, you must select a suitable microcontroller.

A microcontroller is a compact integrated circuit designed for a specific operation. It has an integrated microprocessor, memory, and associated circuits. For the assignment, you are going to use Arduino Uno microcontroller board.



Figure 1 - Arduino

For ease of use, you can use the Arduino Uno development board. However, if your project needed to have additional features, it is acceptable to use any preferable microcontroller (Ex. STM32, ESP32).

To get started with Arduino, you can follow <https://www.arduino.cc/en/Guide> .

You can refer to the following tutorials for further knowledge:

1. Arduino

- <https://www.arduino.cc/en/Tutorial/HomePage>.
- <https://arduino-tutorials.net/>.
- https://www.youtube.com/watch?v=zJ-LqeX_fLU.
- https://www.youtube.com/watch?v=iI_EJvZUT94.
- <https://www.youtube.com/c/Robotszu>.

2. IoT Dashboards

- <https://blynk.io/>
- <https://thingsboard.io/>
- <https://thinger.io/>

3. Basic Electronics

- <https://www.youtube.com/c/Robotszu>
- <https://www.youtube.com/watch?v=eFPTBATfX70list=PLwjK_iyK4LLCAN5TddEZyliChEMpF0oOL>
- <https://www.youtube.com/watch?v=r-X9coYTOV4list=PLah6faXAgguOeMUIxS22ZU4w5nDvCl5gsindex=1>

4. Example Projects

- <https://www.youtube.com/watch?v=OL7TNx9RquE>